

Optoelectronics Introduction

Optoelectronics

光电子学

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AI辅助学习

导入所有课本教材以及参考文献与讲义

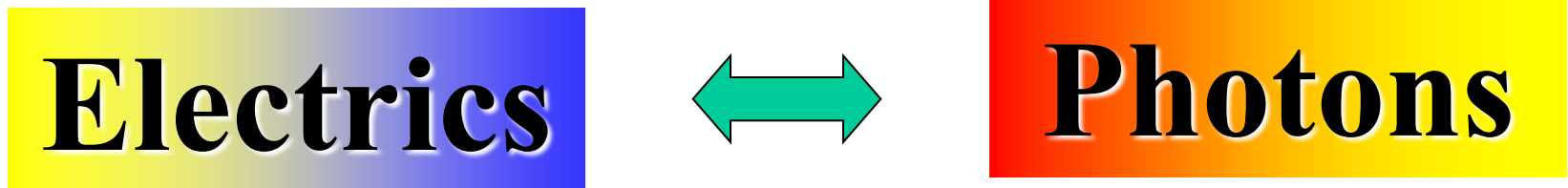
- 该领域手游专家公认的5个核心思维模型是什么？
- 该领域的专家们在哪三个问题上存在根本分歧，分歧点是什么？
- 提出10个问题来区别一个真正理解的人和一个只会死记硬背的人

导入所有课本教材以及参考文献与讲义

- 搭建学习阶梯？
- 该领域的专家们认为哪20%问题是关键核心知识？底层相同的核心逻辑是什么？
- AI考官提出10个问题来区别一个真正理解的人和一个只会死记硬背的人

- 费曼追问法：将自己的理解发给AI，请他按苏格拉底提问，持续深挖追问
- 苏轼八面受敌法：从八个视角提问
- 芒格串联法

Optoelectronics

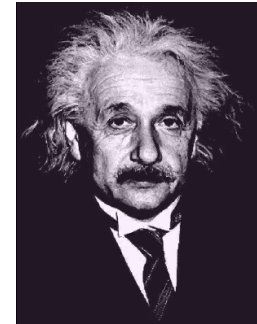


A field of technology that combines the physics of light with electricity.

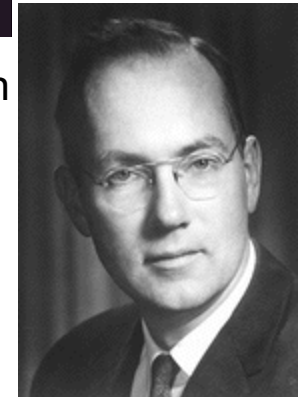
Encompasses the study, design and manufacture of **hardware devices that convert electrical signals into photon signals and vice versa.**

Invention of Laser

- LASER – Light Amplification by Stimulated Emission of Radiation
- 1917
 - Albert Einstein first theorized about the process which makes lasers possible called "Stimulated Emission"
- 1951
 - Charles H. Townes conceived the concept of MASER (Microwave Amplification by Stimulated Emission of Radiation)
- 1954
 - First MASER device by Townes, Gould and Zerger
- 1958
 - Schawlow and Townes theoretically shown that masers could be made in the optical and infrared region
 - "Infrared and Optical Masers," published in the December 1958 *Physical Review*.
 - Received a patent for the invention of the laser in 1960



Albert Einstein



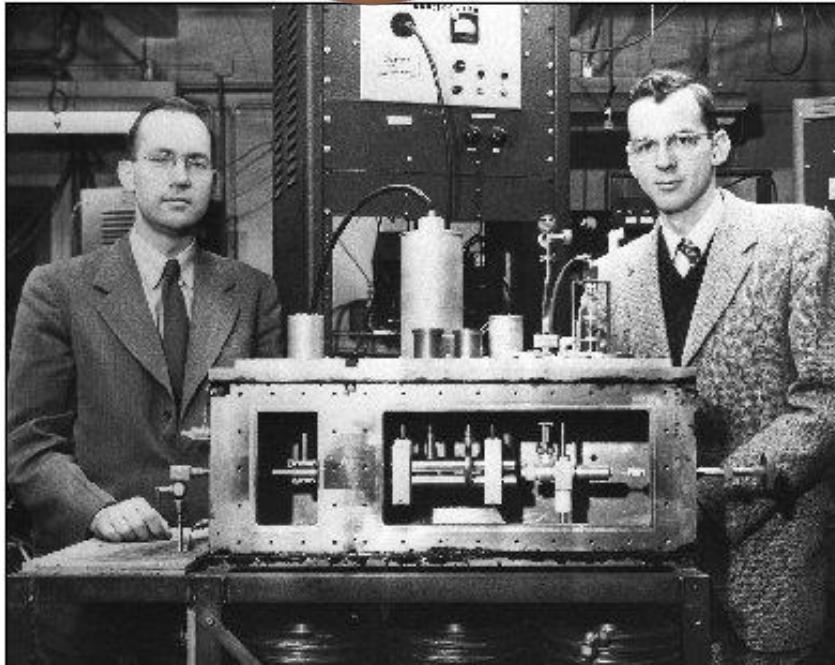
Charles H. Townes



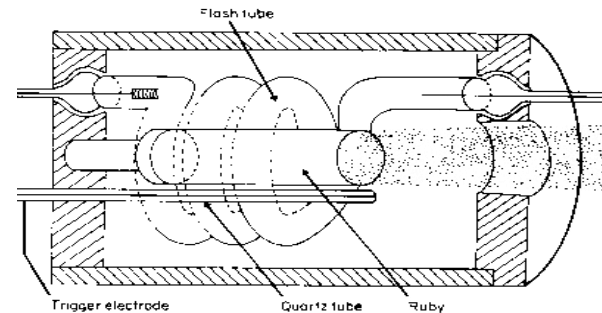
Arthur L. Schawlow



In 1960, Theodore H. Maiman made the first working laser – Ruby laser @ 694nm



Charles Townes (left) and James P. Gordon and their maser, a device that greatly amplifies microwaves



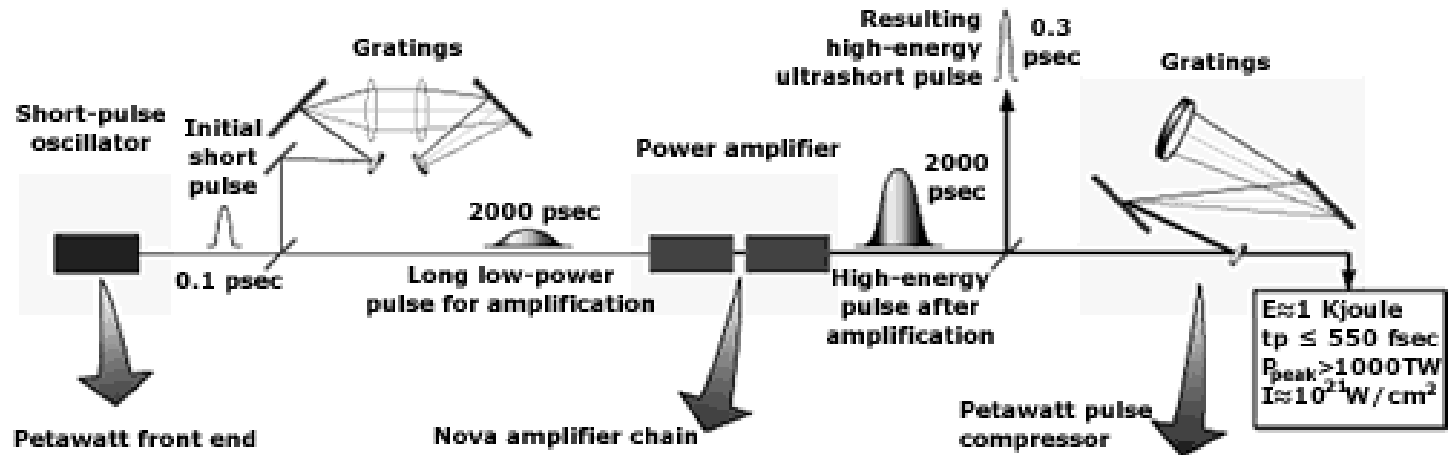
The history of laser in China

1961年中国红宝石激光器（王之江）



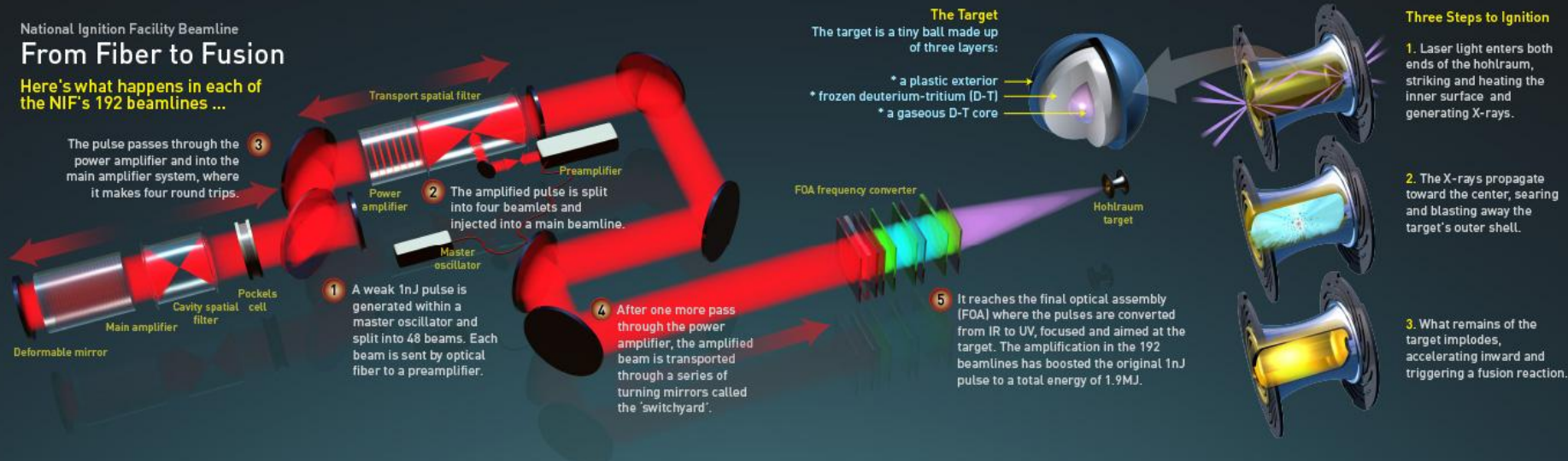
Optoelectronics Applications

Laser fusion (1) (new energy)

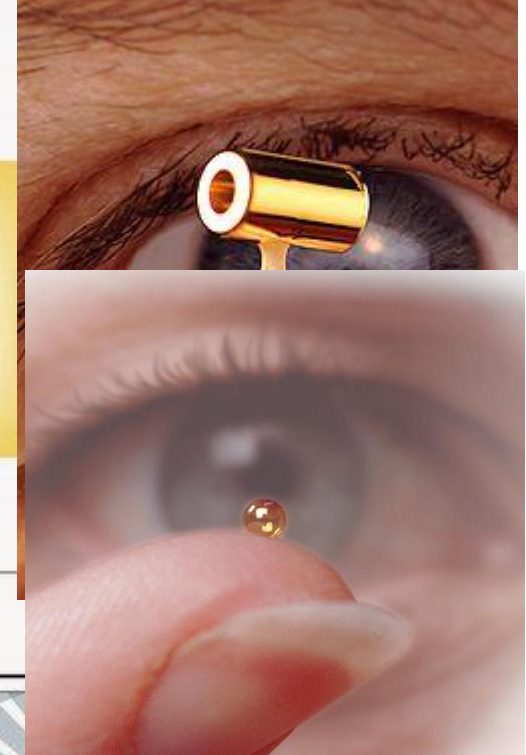
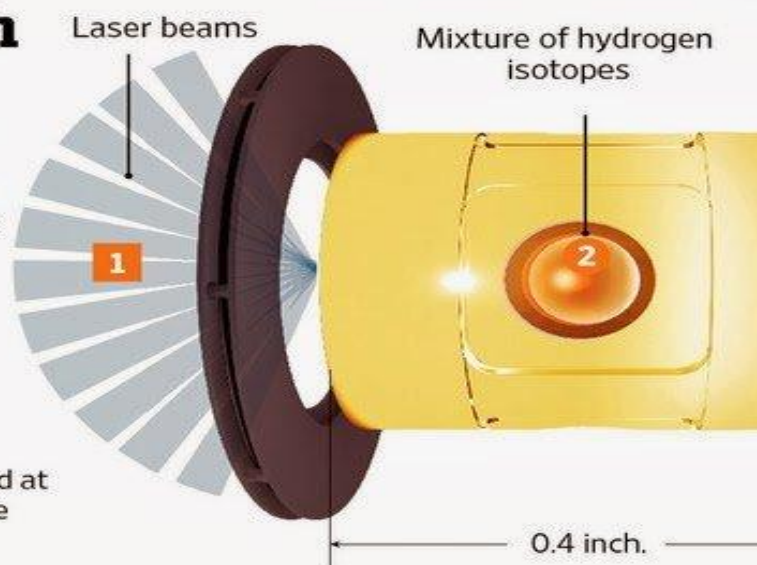
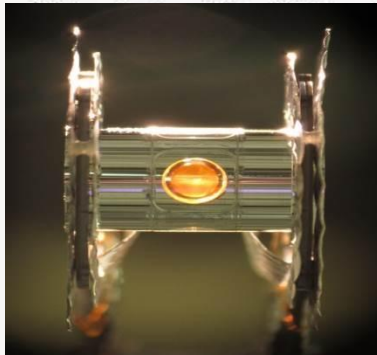


National Ignition Facility Beamline From Fiber to Fusion

Here's what happens in each of the NIF's 192 beamlines ...



Star Search



1 192 laser beams are fired at a tiny gold can about the size of a dime.

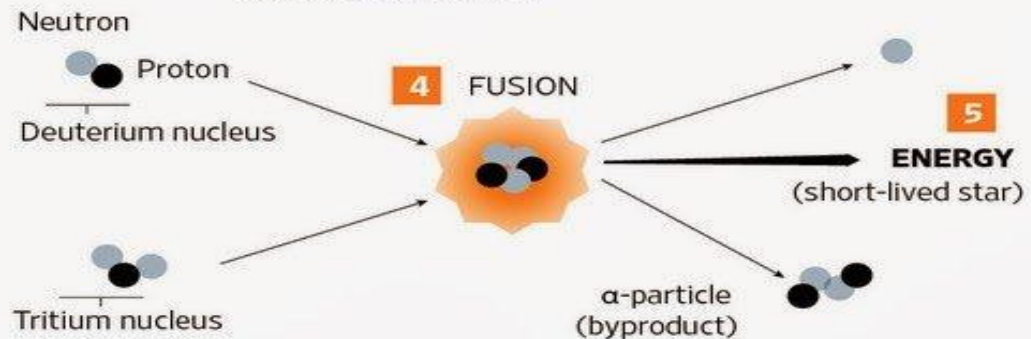
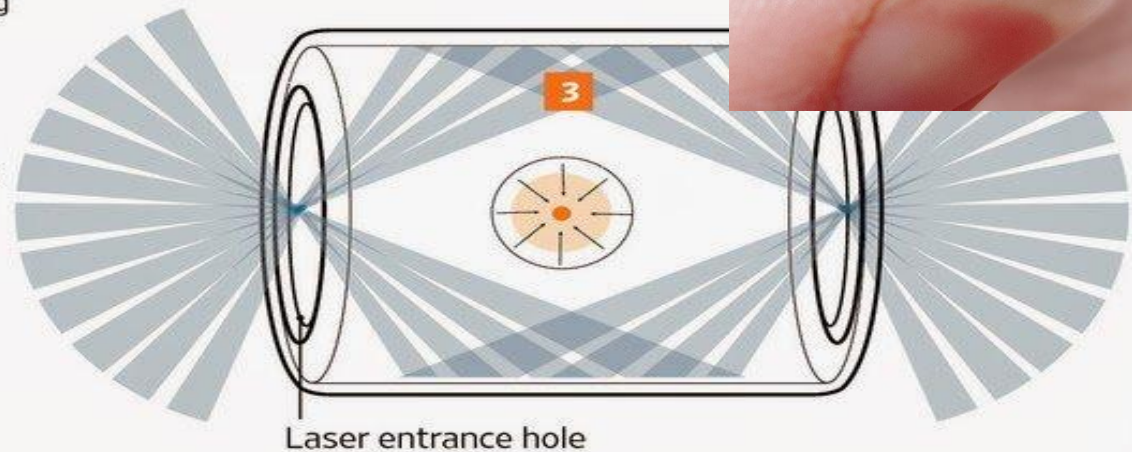
2 Inside the can was a pea-sized capsule containing the fuel—a mixture of two hydrogen isotopes, deuterium and tritium.

3 When laser beams entered the gold can, they struck its inside walls and created X-rays. Those X-rays then bathed the capsule, creating tremendous pressure and crushing the fuel capsule to 1/35th of its radial size.

4 As the capsule shrank, the fuel rapidly converged around the hollow center, which generated even higher pressures and temperatures, triggering a fusion reaction.

5 To achieve ignition—the key to practical fusion energy—the reaction must become self-sustaining, but that is still a long way off.

Source: Nature



Laser machine

Lasing Materials

CO₂

Nd

Nd-YAG

Fiber laser

Applications

Boring Cutting/Scribing Engraving

High-energy pulses Low repetition speed (1 kHz)

Boring

Very high energy pulses Boring Engraving Trimming

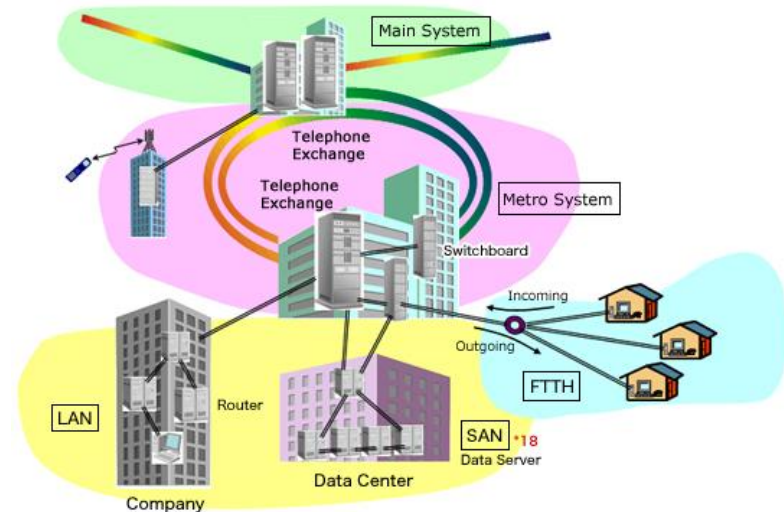
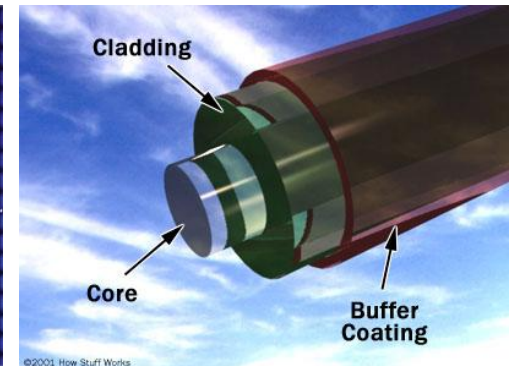
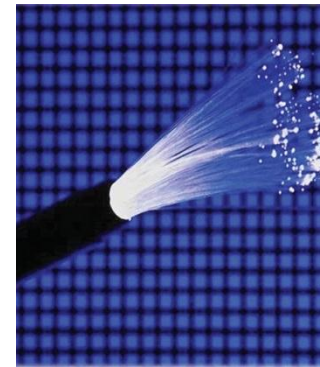
Convenience in the application



2009 Nobel Prize

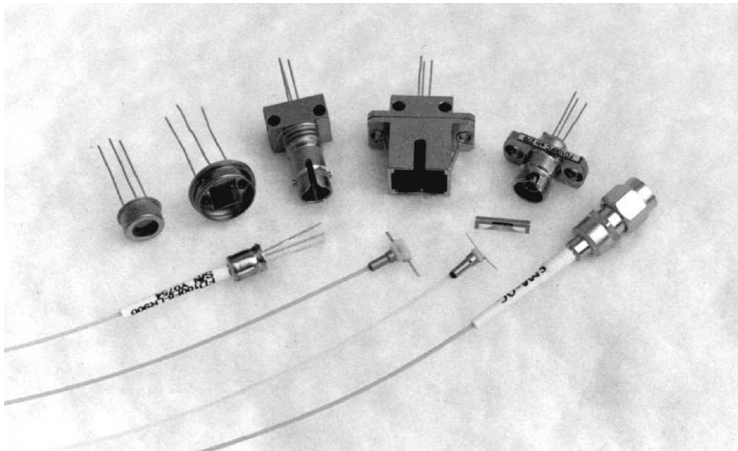
For groundbreaking achievements concerning the transmission of light in fibers for optical communication

1966 Charles K. Kao



InGaAs based photodetectors

Optical Communication



InGaAs based photodetectors is widely used in optical communication system, including fiber-pigtailed photodiodes, Amplifier etc.

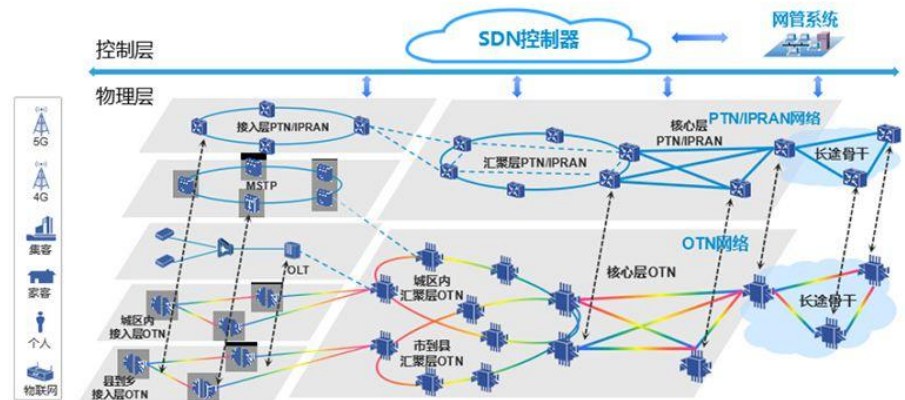


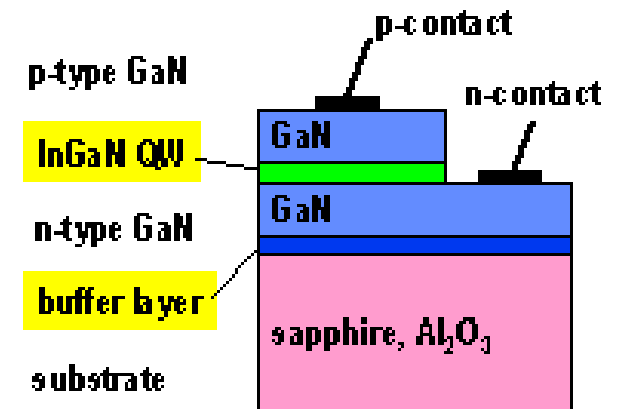
图1 面向未来的OTN网络架构

Light sources

- Incandescent lamp
- Gas discharge lamp
- Semiconductor light source



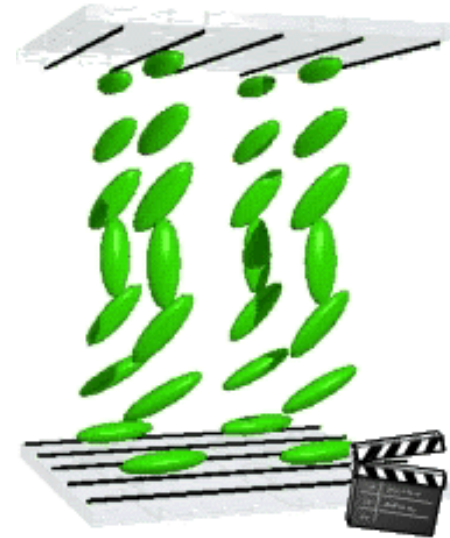
2014 Nobel Price (Physics)



Electro-optical effect (Display)

- Electro-optical modulation technique
 - EO crystal
 - Liquid crystal

KDP crystal



Liquid Crystal and Liquid crystal display

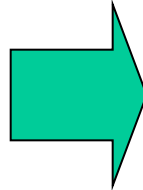
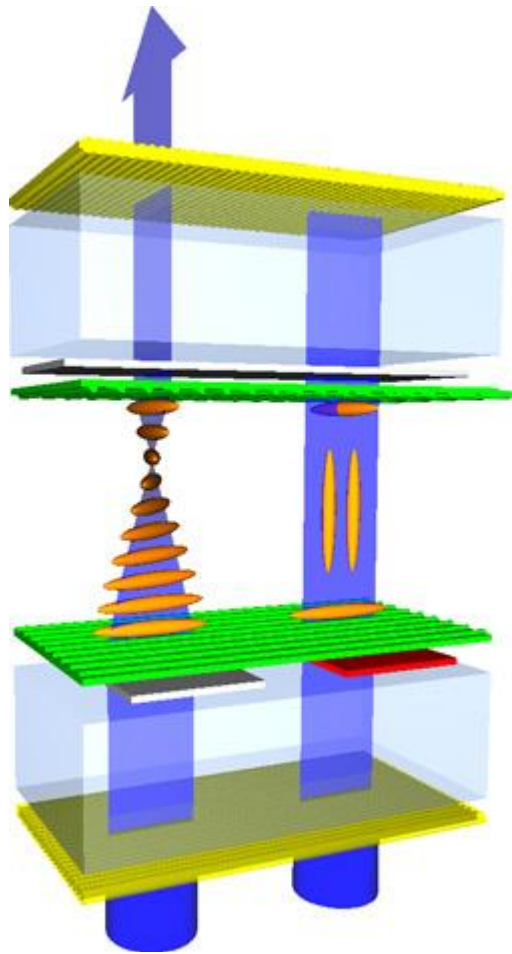
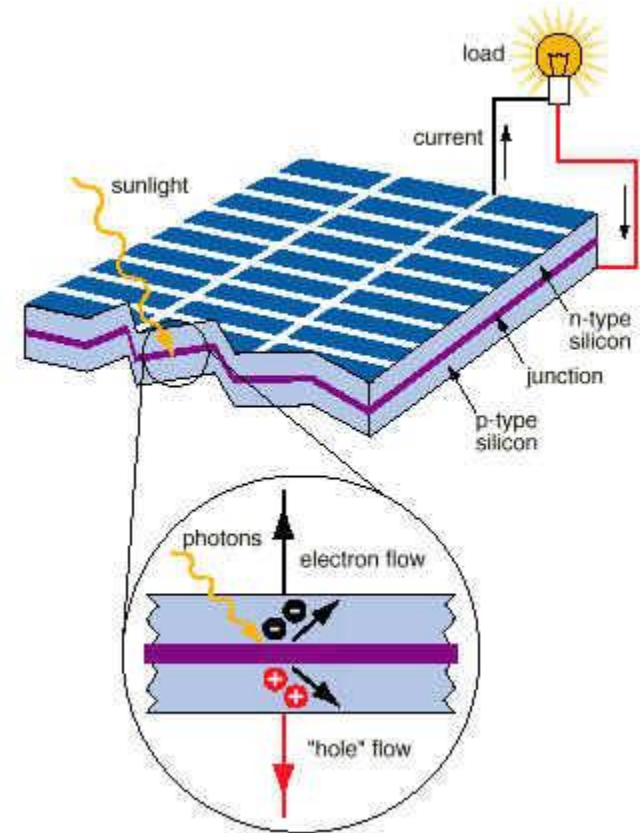


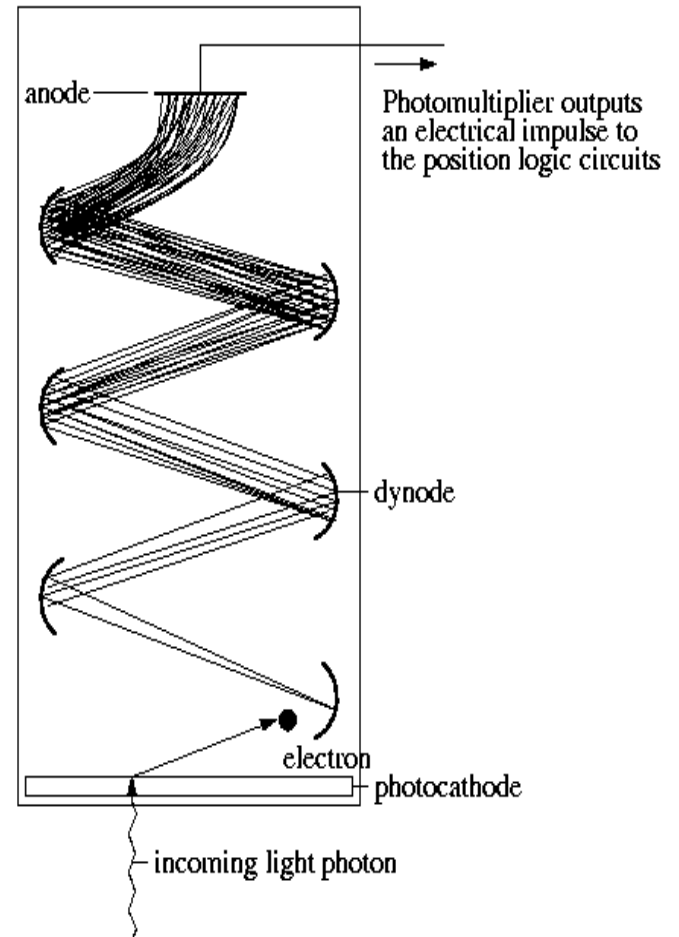
Photo detector

- Photovoltaic effect — — Solar cell



Photodetector (2)

- UV+visible+nIR
- PMT Photomultiplier tube
(vacuum + photocathode)



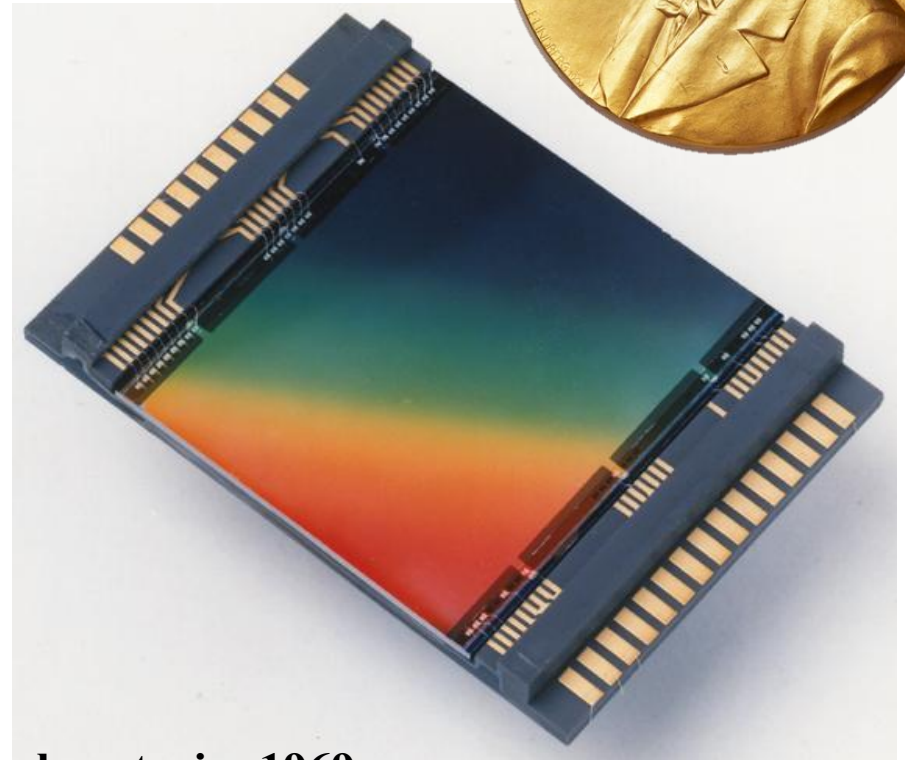
A Photomultiplier Tube

photodetector (2)



- CCD device

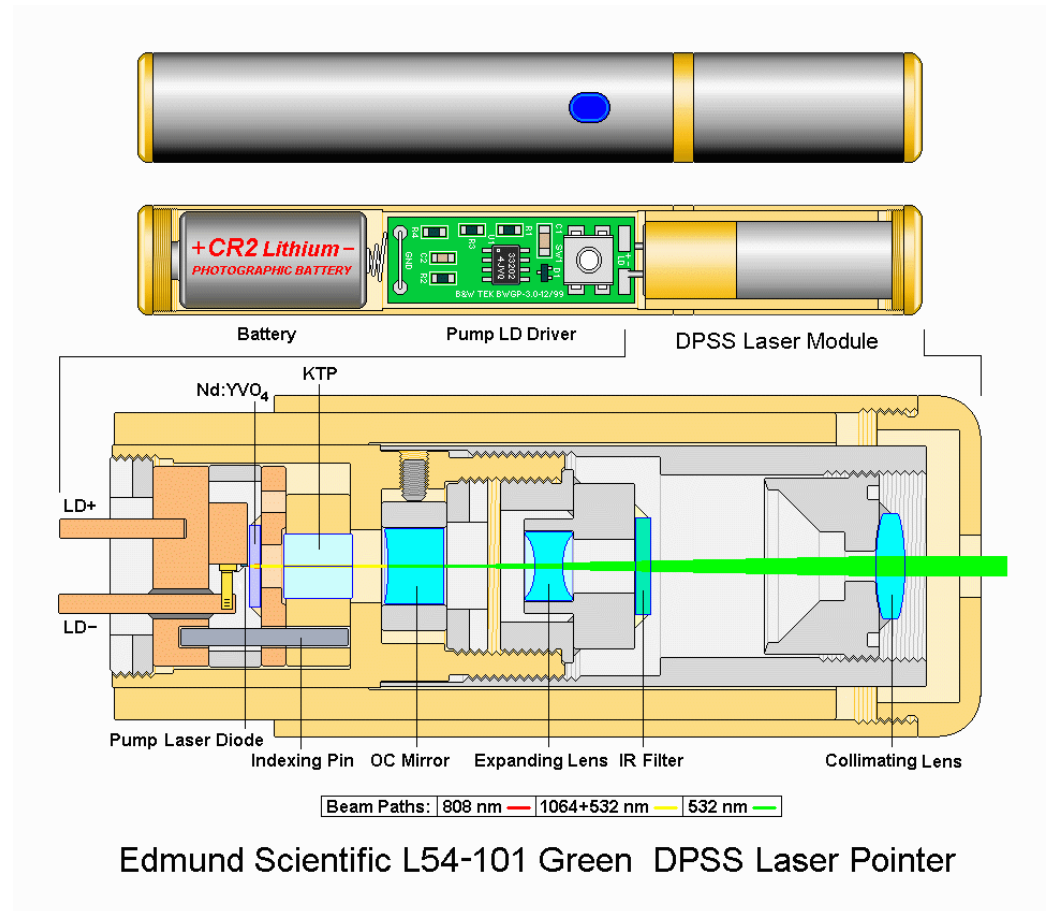
2009 Nobel Prize



Willard S. Boyle George E. Smith Bell Laboratories 1969
for the invention of an imaging semiconductor circuit –
the CCD sensor"

Nonlinear optics and crystal

- Double frequency
- 3rd frequency
- We can almost get all the laser in any wavelength we like.
- Laser pointer



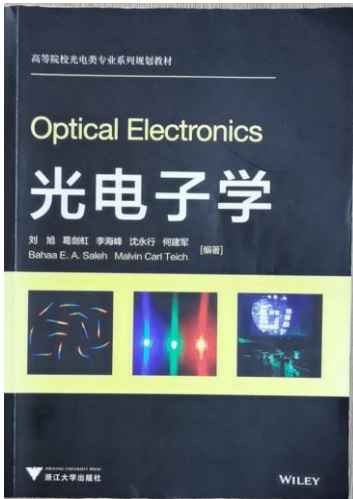
The OE's industries

- Laser industry
- Optical communication
- Display industry
- Imaging Industry (2D & 3D imaging: Lidar)
- Photo-detectors (Solar cell) industry
- Lighting industry
- Optical computing

Optoelectronics (contents)

- Laser theory and techniques
- Semiconductor Photonics device
- Photodetectors
- Modulation
- Nonlinear optics

Text books and Reference



Text book

- H. Salsh: Fundamental of photonics 2007
- A.E. Siegman: Lasers (University Science Books, 1986)
- A. N. Oraevskiy, Gaussian Beams and Optical Resonators, Nova Science, 1996.
- 阎吉祥, 光电子学, 清华大学出版社, 2017
- 周炳琨, 激光原理, 国防工业出版社, 2014
- Robert W. Boyd Nonlinear optics
- Lukas Chrostowski, Michael Hochberg Silicon Photonics Design: From Devices to Systems, Cambridge University Press

